Econometrics

Sample final exam

Instructor: Dr. Doron Israeli

<u>General instructions</u>:

You have 3.5 hours to solve this exam.

Work on this exam is to be done individually.

You may use your notes and the R examples presented in class.

When submitting your solutions, be sure to upload to provide soft copies of:

1. A *.doc file named "**FinalExam_ID.doc**", which comprises your solutions to relevant questions (ID indicates your identification number). The *.doc file should contain detailed solutions to the questions, including the relevant R code and the output.

2. An R script named "**FinalExam_ID.R**", which comprises the entire R program you used to solve the questions. The R script should
   - be ready to run as it is and generate the results you report in the *.doc file.
   - be properly structured, e.g., indicate the number of question before each sequence of commands, and provide comments to facilitate understanding.

Grading scheme: Part A is worth 35 points and Part B is worth 65 points.

<u>Questions</u>:

The goal of this exercise is to examine the claim that during the time period from July 1992 to June 1993, the well-known relation between average future stock returns (i.e., AvgMonRet) and observed levels of book-to-market ratio (i.e., a firm's BTM ratio) was concentrated among firms that do not belong to "Computers" industry (according to the BBHL industry classification). In other words, the Fama and French (1992)[1] discovery does not apply to firms from the "Computers" industry.

To perform the examination you need to use the Signals_A_62016.csv data set, i.e., a panel data set that comprises a select number of trading signals as well as stock returns of firms whose stock is traded on major US exchanges.
The DataSets_Descriptions.pdf file provides detailed descriptions of all variables included in these data sets.

<u>Part A (35 points)</u> Loading the data set into R, creating and describing the variables of interest, and setting up the CrossSecDF data set for future analyses:

1. Load the Signals_A_62016.csv data set into R, define the PFormDate variable as a date variable, and add to the Signals_A_62016.csv data set an indicator (dummy) variable, "CompInd", that takes on the value of 1 if an observation belongs to the "Computers" industry, according to BBHL industry classification, and 0 otherwise.
   - How many observations, unique firms (i.e., PermNo), trading portfolio formation dates (i.e., PFormDate), and BBHL industries does the Signals_A_62016.csv comprise?
   - How many observations and how many unique firms belong to the "Computers" industry?

2. Create a new cross-section data set, CrossSecDF, that comprises observations for which the trading portfolio formation date (i.e., PFormDate) is June 1992. Make sure that CrossSecDF contains only observations with non-missing values of Size, BTM, ETP, and AvgMonRet.
   - How many observations and unique firms from "Computers" industry and how many observations and unique firms from other industries does the CrossSecDF data set comprise?

<u>Part B (65 points)</u> Regression analyses using the CrossSecDF data set, i.e., a cross-section of firms with portfolio formation date (i.e., PFormDate) as of end of June, 1992.

One way to examine whether during July 1992-June 1993 the relation between BTM and AvgMonRet was concentrated among firms that do not belong to "Computers" industry is to estimate the following regression models using OLS.

[1] Fama, E. F., and K. R. French, 1992. The cross-section of expected stock returns. *Journal of Finance* 47: 427-465.

<u>Equation 1</u>: $AvgMonRet = \boldsymbol{\beta_0} + \boldsymbol{\beta_1}\log(Size) + \boldsymbol{\beta_2}\log(BTM) + \boldsymbol{\beta_3}ETP + U$

<u>Equation 2</u>:

$$AvgMonRet = \boldsymbol{\beta_0} + \boldsymbol{\beta_1}\log(Size) + \boldsymbol{\beta_2}\log(BTM) + \boldsymbol{\beta_3}ETP + \boldsymbol{\beta_4}CompInd + \boldsymbol{\beta_5}CompInd \times \log(BTM) + U$$

Where;

$AvgMonRet$, $Size$, BTM, and ETP are defined in the DataSets_Descriptions.pdf file;

CompInd is an indicator variable that takes on the value of 1 if an observation belongs to the "Computers" industry, according to BBHL industry classification, and 0 otherwise.

log( ) denotes the natural logarithm.

3. Estimate Equation 1 using the <u>combined sample of firms</u> and <u>separately</u> for firms that belong or do not belong to "Computers" industry (i.e., 3 estimations in total).
   - For each of the three estimations, report and interpret the point as well as two-tailed 95% interval estimates of $\boldsymbol{\beta_2}$.
     - Which standard errors do you use in making your inferences? Why?
     - Based on these estimates, would you agree with the claim? Explain!
   - What is the downside of exploring the claim using separate estimates of Equation 1?
   - **(Bonus question)** Compute two-tailed 95% interval estimate for the difference between skewness of errors for firms that belong to "Computers" and firms that belong to other industries.
     - Which method do you use to compute the interval estimate? Why? [**Hint**: In case you rely on the Bootstrap approach, use 10000 Bootstrap samples)
     - What do you learn from the interval estimate?

4. Estimate Equation 2 using the combined sample of firms (i.e., 1 estimation).
   - Report and interpret the point estimates of $\boldsymbol{\beta_2}$, $\boldsymbol{\beta_4}$, and $\boldsymbol{\beta_5}$.
   - Using a significance level of 5%, test the hypothesis that in the population of interest, $\beta_2 = 0$. What do you learn from this test?
   - Using a significance level of 5%, test the hypothesis that in the population of interest, $\beta_2 + \beta_5 = 0$. What do you learn from this test?
     - Which standard errors do you use? Why?
   - Based on the estimates of Equitation 2, would you agree with the claim? Why?

5. You classmate argues that another way to examine the claim would be to estimate the following regression model by OLS. The idea is to test whether, in Equation 3, β_3 is significantly different from 0.

<u>**Equation 3**</u>**:**

$$\boldsymbol{CompInd = \beta_0 + \beta_1 \log(Size) + \beta_2 \log(BTM) + \beta_3 ETP + \beta_4 AvgMonRet + U}$$

- Do you agree with your classmate? Explain!

- Estimate Equation 3 using OLS.
  - Report and interpret the coefficient estimate on $AvgMonRet$.

- Re-write Equation 3 as a Logit model and estimate it using MLE.
  - Report and interpret the coefficient estimate on $AvgMonRet$.

- Compare the marginal effect of $AvgMonRet$ on $CompInd$ as it is reflected from Equation 3 when you estimate it by OLS and when you frame it as a Logit model and estimate it using MLE.
  [**Hint**: When computing the marginal effect in the Logit model, you are expected to use the average values of the independent variables]

  - What do you learn from the comparison?

GOOD LUCK!!